

Class 10 Task Program:- ADC PWM Modular Programming

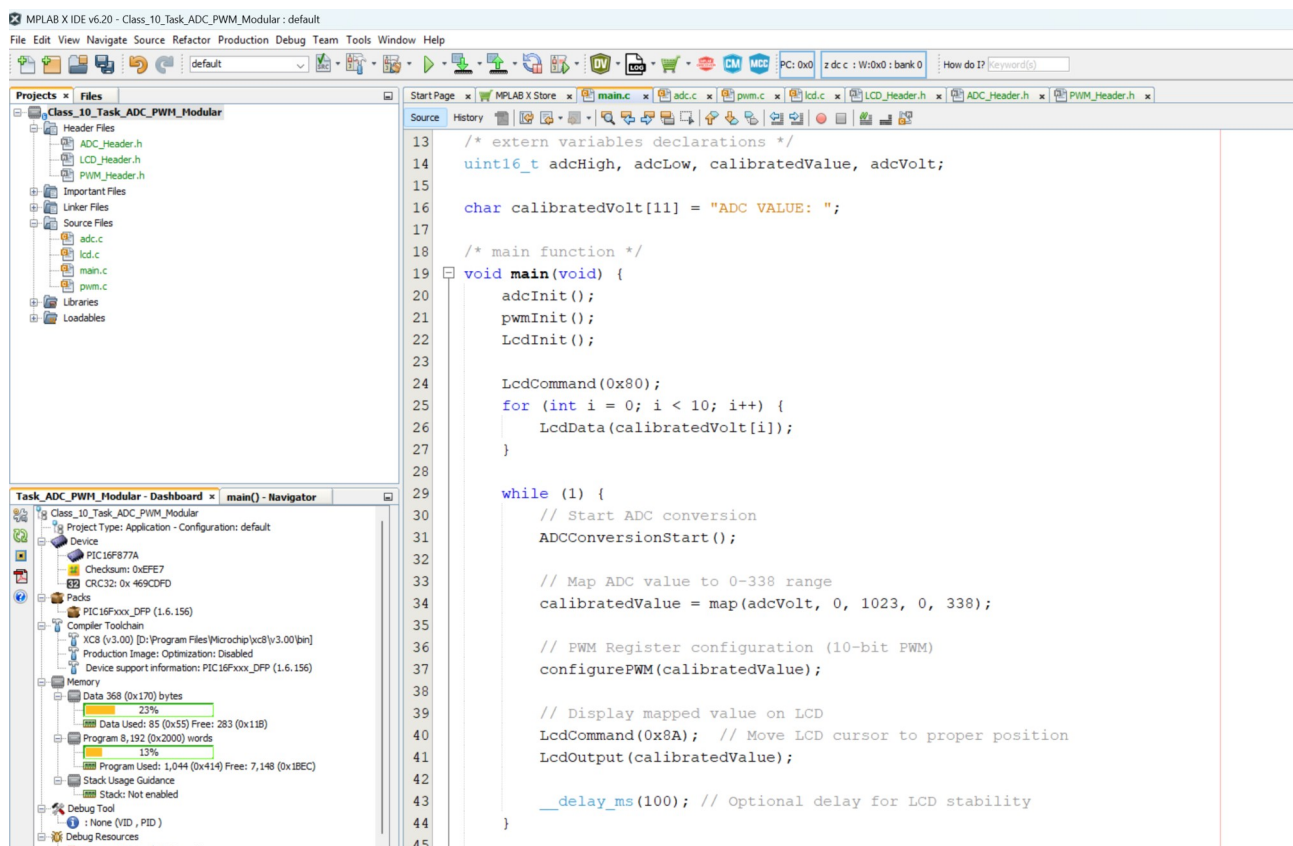


Figure 1: modular Programming Setup

```

/*
 * File: main.c
 * Author: sagar
 *
 * Created on 2 June, 2025, 10:37 PM
 */
#include <xc.h>
#include "ADC_Header.h"
#include "LCD_Header.h"
#include "PWM_Header.h"
#define _XTAL_FREQ 6000000

/* extern variables declarations */
uint16_t adcHigh, adcLow, calibratedValue, adcVolt;

char calibratedVolt[11] = "ADC VALUE: ";

/* main function */
void main(void) {
    adcInit();
    pwmInit();
    LcdInit();

    LcdCommand(0x80);
    for (int i = 0; i < 10; i++) {
        LcdData(calibratedVolt[i]);
    }

    while (1) {
        // Start ADC conversion
        ADCCConversionStart();

        // Map ADC value to 0-338 range
        calibratedValue = map(adcVolt, 0, 1023, 0, 338);

        // PWM Register configuration (10-bit PWM)
        configurePWM(calibratedValue);

        // Display mapped value on LCD
        LcdCommand(0x8A); // Move LCD cursor to proper position
        LcdOutput(calibratedValue);

        __delay_ms(100); // Optional delay for LCD stability
    }

    return;
}

```

```

1  /*
2  * File:    pwm.c
3  * Author:  sagar
4  *
5  * Created on 2 June, 2025, 10:58 PM
6  */
7
8  #include<xc.h>
9  #include "PWM_Header.h"
10 #include <stdint.h>
11 #define _XTAL_FREQ 6000000
12
13 void pwmInit(void) {
14     TRISC = 0xFB; // Make RC2 (CCP1) an output
15
16     CCP1CON = 0x0C; // PWM mode, 10-bit
17     T2CON = 0x06; // Timer2 ON, prescaler 1:16
18     PR2 = 0x5E; // Set period
19
20     TMR2 = 0; // Reset Timer2 counter
21     TMR2ON = 1; // Turn on Timer2
22 }
23
24 void configurePWM(uint16_t calibratedValue)
25 {
26     CCP1RL = (uint8_t)(calibratedValue >> 2); // Upper
27     uint8_t lowerBits = (uint8_t)((calibratedValue & 0x0F) << 2);
28     CCP1CON = (CCP1CON & 0xCF) | lowerBits;
29 }
30
31
32

```

```

1  /*
2  * File:    PWM_Header.h
3  * Author:  sagar
4  *
5  * Created on 2 June, 2025, 10:58 PM
6  */
7
8  #ifndef PWM_HEADER_H
9  #define PWM_HEADER_H
10
11 #ifdef __cplusplus
12 extern "C" {
13 #endif
14
15 void pwmInit(void);
16 void configurePWM(uint16_t calibratedValue);
17
18 #ifdef __cplusplus
19 }
20 #endif
21
22 #endif /* PWM_HEADER_H */
23
24
25
26
27
28
29

```

Figure 2: pwm;- .c .h

```

1  /*
2  * File:    lcd.c
3  * Author:  sagar
4  *
5  * Created on 2 June, 2025, 10:48 PM
6  */
7
8
9  #include<xc.h>
10 #include "LCD_Header.h"
11
12 #define _XTAL_FREQ 6000000
13
14
15 /* LCD init function */
16 void LcdInit(void) {...21 lines}
17
18
19 /* LCD output function */
20 void LcdOutput(uint16_t i) {...11 lines}
21
22
23 /* Function to send data (characters) to the LCD */
24 void LcdData(uint8_t i) {...7 lines}
25
26
27 /* Function to send commands to the LCD */
28 void LcdCommand(uint8_t i) {...7 lines}
29
30
31

```

```

1  /*
2  * File:    LCD_Header.h
3  * Author:  sagar
4  *
5  * Created on 2 June, 2025, 10:48 PM
6  */
7
8  #ifndef LCD_HEADER_H
9  #define LCD_HEADER_H
10
11 #ifdef __cplusplus
12 extern "C" {
13 #endif
14
15 /* Function prototypes */
16 void LcdInit(void);
17 void LcdOutput(uint16_t i);
18 void LcdData(uint8_t i);
19 void LcdCommand(uint8_t i);
20
21 #ifdef __cplusplus
22 }
23 #endif
24
25 #endif /* LCD_HEADER_H */
26
27

```

Figure 3: LCD:- .c .h

```
1 /*
2  * File:   adc.c
3  * Author: sagar
4  *
5  * Created on 2 June, 2025, 10:53 PM
6  */
7
8 #include <xc.h>
9 #include "ADC_Header.h"
10
11 #define _XTAL_FREQ 6000000
12
13 extern uint16_t adcVolt;
14 extern uint16_t adcLow;
15 extern uint16_t adcHigh;
16
17 void adcInit(void) {
18     /****** ADC Init *****/
19     ADCON1 = 0x8E;
20     ADCON0 = 0x81;
21     __delay_ms(10);
22 }
23
24 void ADCCConversionStart(void) {
25     ADCON0 |= 0x04;
26     while (ADCON0 & 0x04) {} // Wait for conversion
27
28     // Read ADC result (10-bit value)
29     adcHigh = ADRESH;
30     adcLow = ADRESL;
31     adcVolt = ((adcHigh << 8) | adcLow); // Right justify
32 }
33
```

```
1 /*
2  * File:   ADC_Header.h
3  * Author: sagar
4  *
5  * Created on 2 June, 2025, 10:53 PM
6  */
7
8 #ifndef ADC_HEADER_H
9 #define ADC_HEADER_H
10
11 #ifdef __cplusplus
12 extern "C" {
13 #endif
14
15 void adcInit(void);
16 void ADCCConversionStart(void);
17 uint16_t map(long x, long in_min, long in_max, long out_min,
18             long out_max);
19
20 #ifdef __cplusplus
21 }
22 #endif
23
24 #endif /* ADC_HEADER_H */
25
26
27
```

Figure 4: ADC:- .c .h